CREDIT RISK CASE STUDY

An approval policy that cuts losses and keeps 91% of volume

Default model, profit-driven cutoff and a policy simulator, built on 589k resolved Lending Club loans

Case study: Northvale Credit is a fictional lender. Built on public Lending Club data for learning purposes.

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THE PROBLEM

Default rates rose from 12.3% to 14.9% between the 2013 and 2015 vintages

The credit committee asked for an approval policy that cuts expected losses without killing loan volume.

14.9%

2015 vintage default rate

36-month loans, up +2.6 pp from 12.3% in 2013

+3.1 pp

from risk within grades

Grade mix alone moved it -0.6 pp; loans within each grade got riskier

19.4%

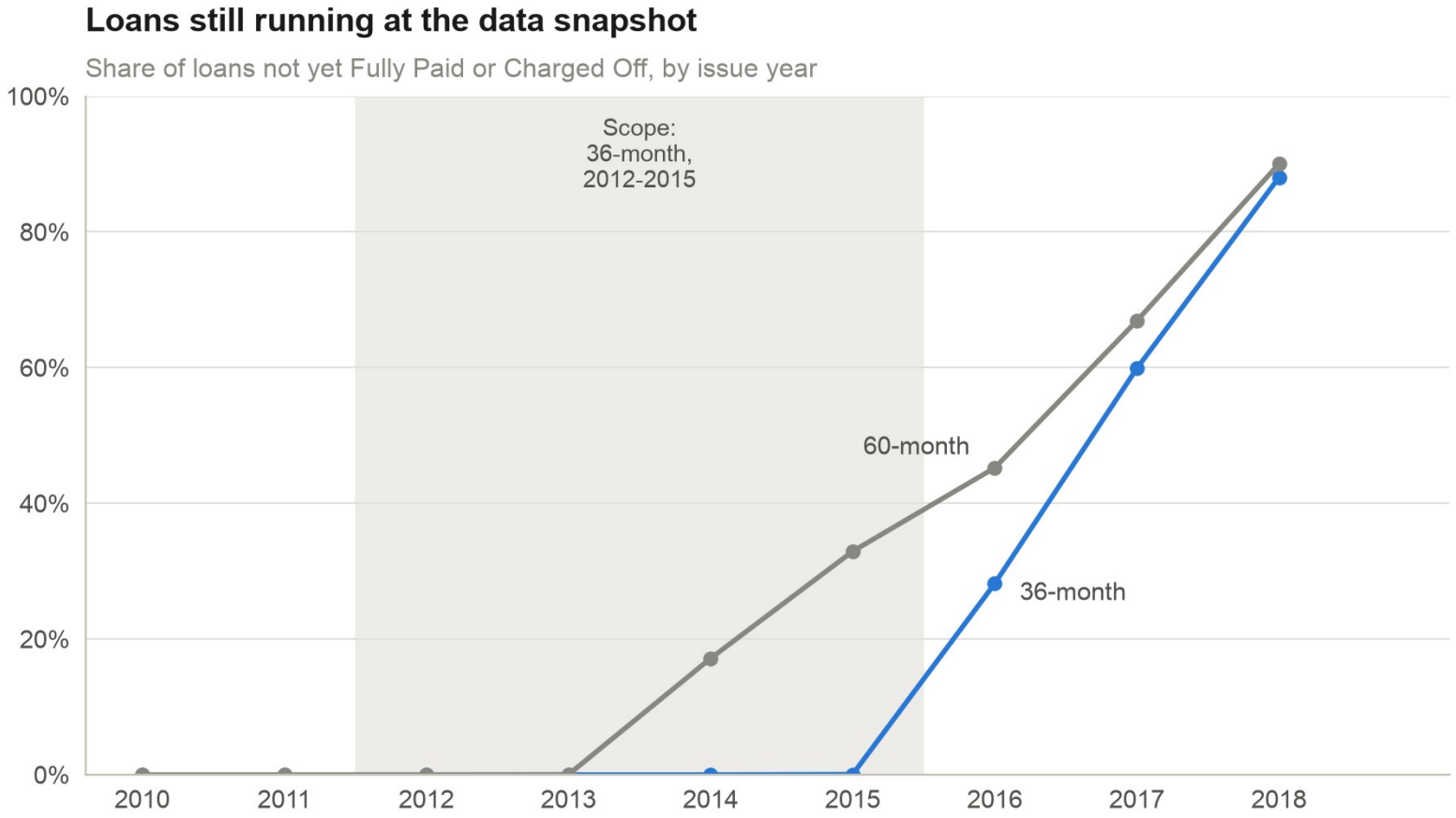
grade C default, 2015

Up from 15.3% in 2013 while its average rate fell from 15.4% to 13.2%

Resolved 36-month loans: 100,422 issued in 2013, 282,787 in 2015. Default = Charged Off or Default.

THE DATA

Only 0.02% of 36-month 2012-2015 loans are unresolved, so defaults are not censored



Source: Lending Club accepted loans 2007-2018Q4, all terms

589k

resolved loans, $7.4B issued 2012-2015

- 28% of 2016 loans still running: keeping them biases defaults
- 60-month loans excluded: 33% of 2015 still running
- Train 2012-14, tune on 2015 H1, test once on 2015 H2

Source: Lending Club accepted loans 2007-2018Q4

APPROACH

Audit the inputs, beat the incumbent grade, then let profit choose the cutoff

1

Scope

36-month loans 2012-2015 with final outcomes, split by issue month, never at random

2

Audit

All 151 columns reviewed; post-origination, pricing and proxy fields removed

3

Model

LC sub-grade benchmark vs WoE scorecard vs LightGBM, recalibrated on 2015 H1

4

Explain

SHAP drivers turned into plain-English reason codes for each decision

5

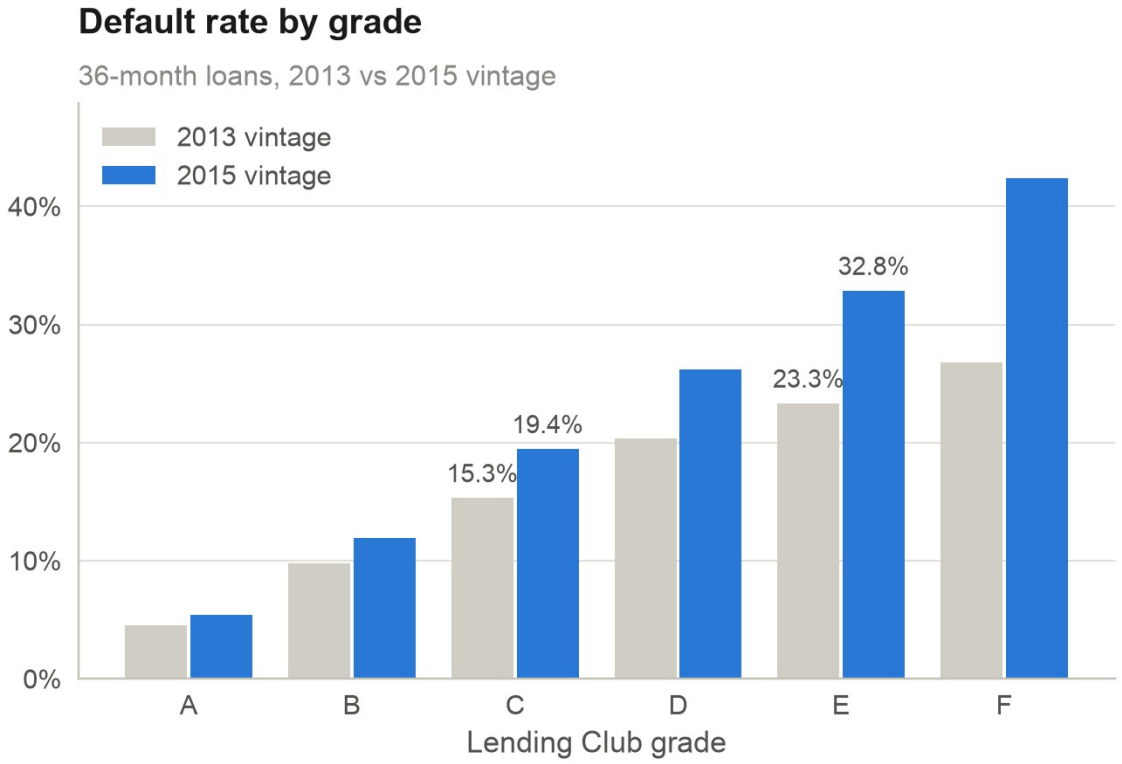
Decide

Profit per loan from PD, price and loss; cutoff picked on 2015 H1, tested on 2015 H2

Fixed seed 42. The cutoff is never tuned on the test window.

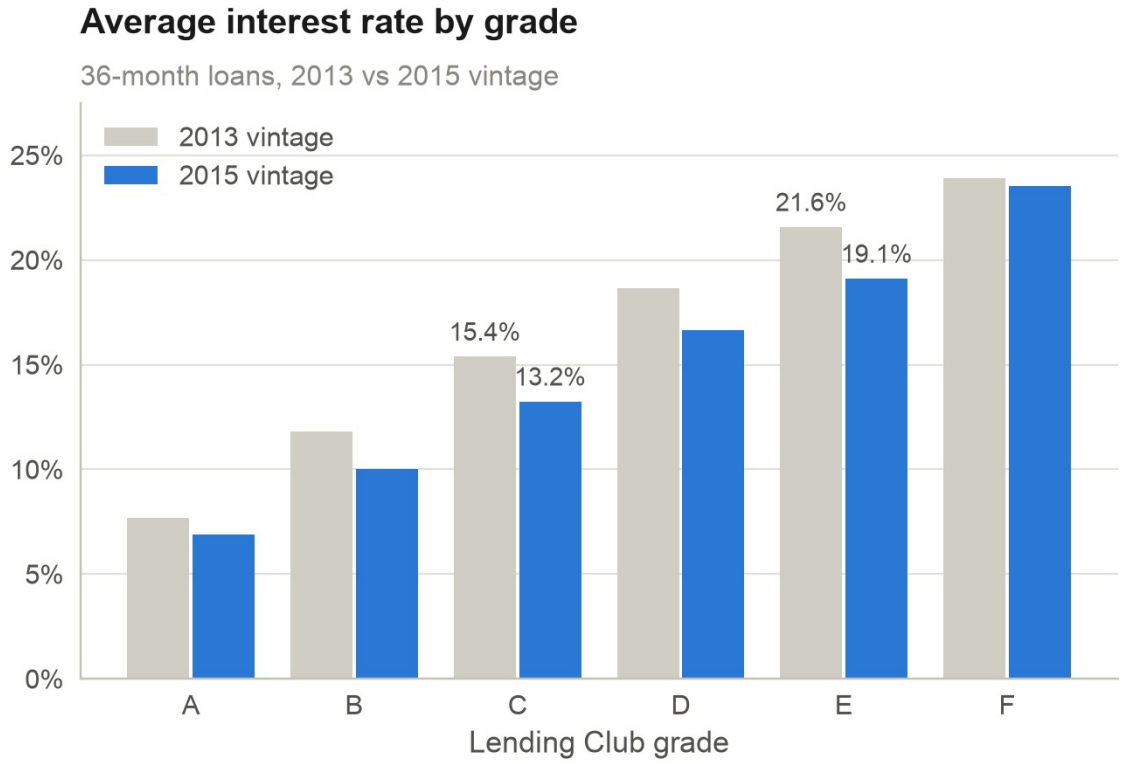
FINDING 1

Grade C loans defaulted 19.4% in 2015 against 15.3% in 2013, at a lower price



Source: Lending Club, 36-month loans; grade G omitted (15 loans in 2013)

Default rates rose inside every grade from A to F.

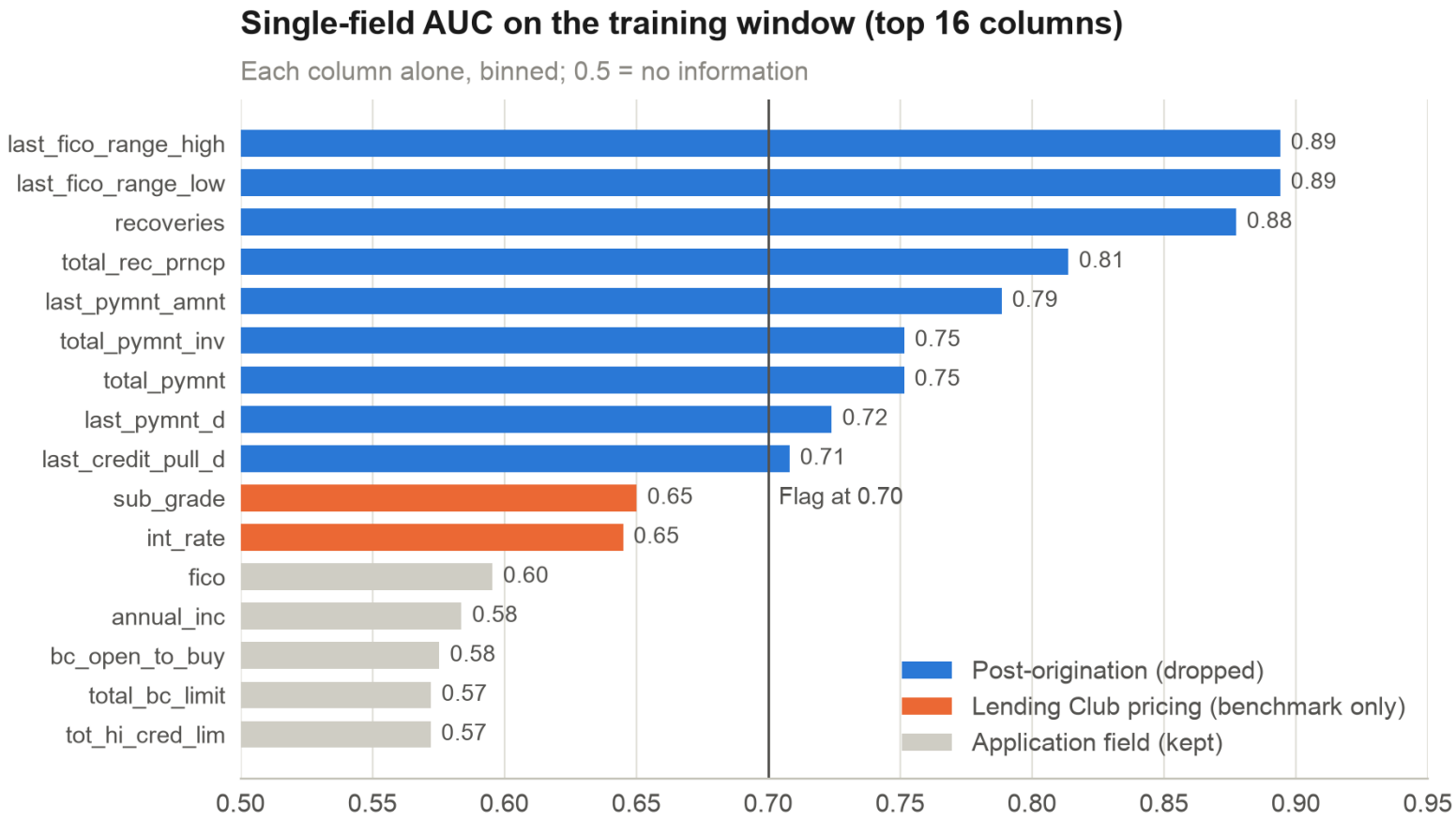


Source: Lending Club, 36-month loans; grade G omitted (15 loans in 2013)

Interest rates fell in grades A to E over the same period.

FINDING 2

Fields recorded after issuance would lift test AUC from 0.70 to 0.95



Source: Lending Club, 36-month loans issued 2012-2014 (training window)

0.95

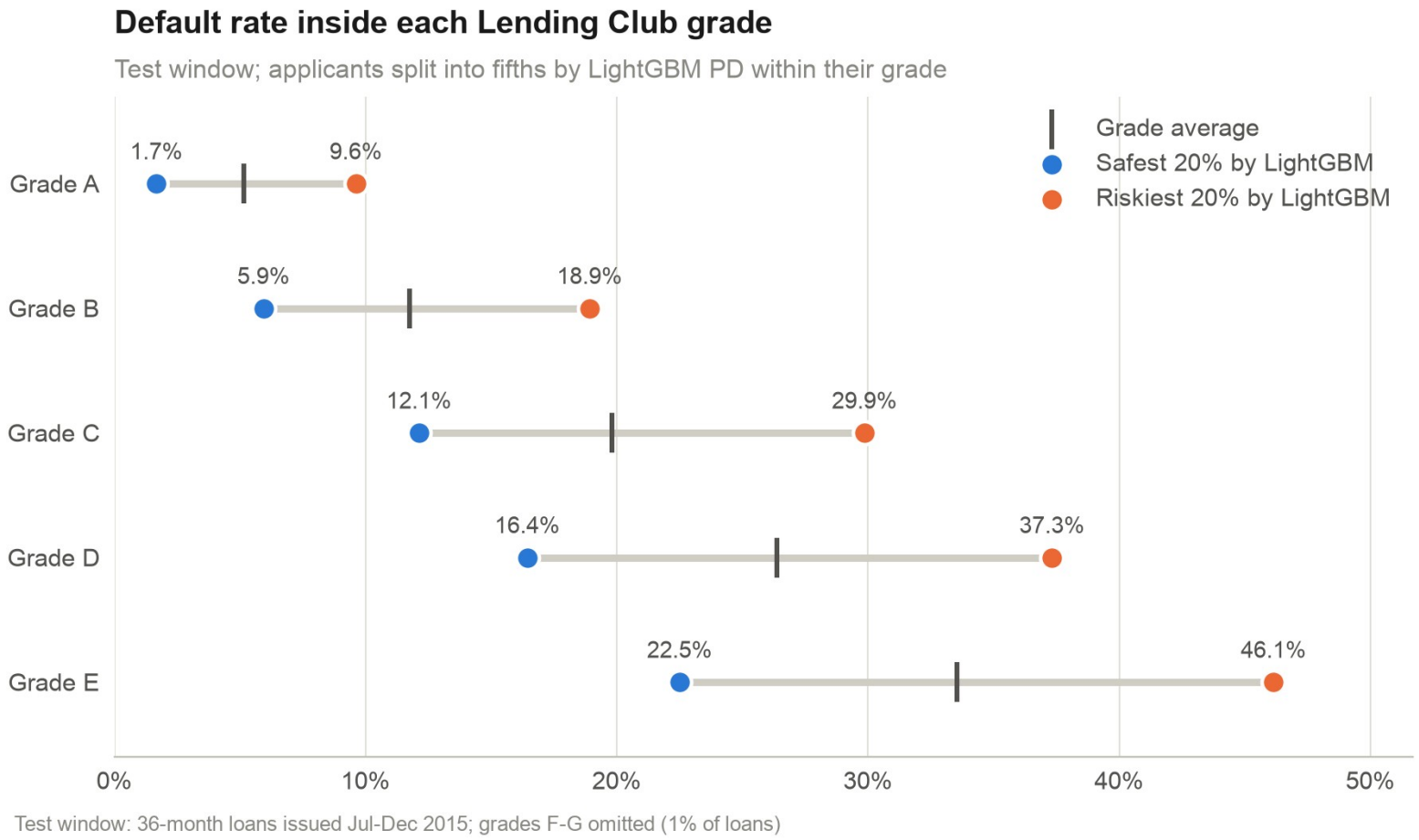
test AUC with refreshed FICO, vs 0.70 without

- 9 columns score above 0.70 alone: all post-origination
- Best real application field: FICO at 0.60
- 38 post-origination columns dropped

Audit of 151 columns: docs/feature_audit.csv

FINDING 3

LightGBM separates 12.1% from 29.9% default rates inside grade C



Test window: 36-month loans issued Jul-Dec 2015; grades F-G omitted (1% of loans)

0.701

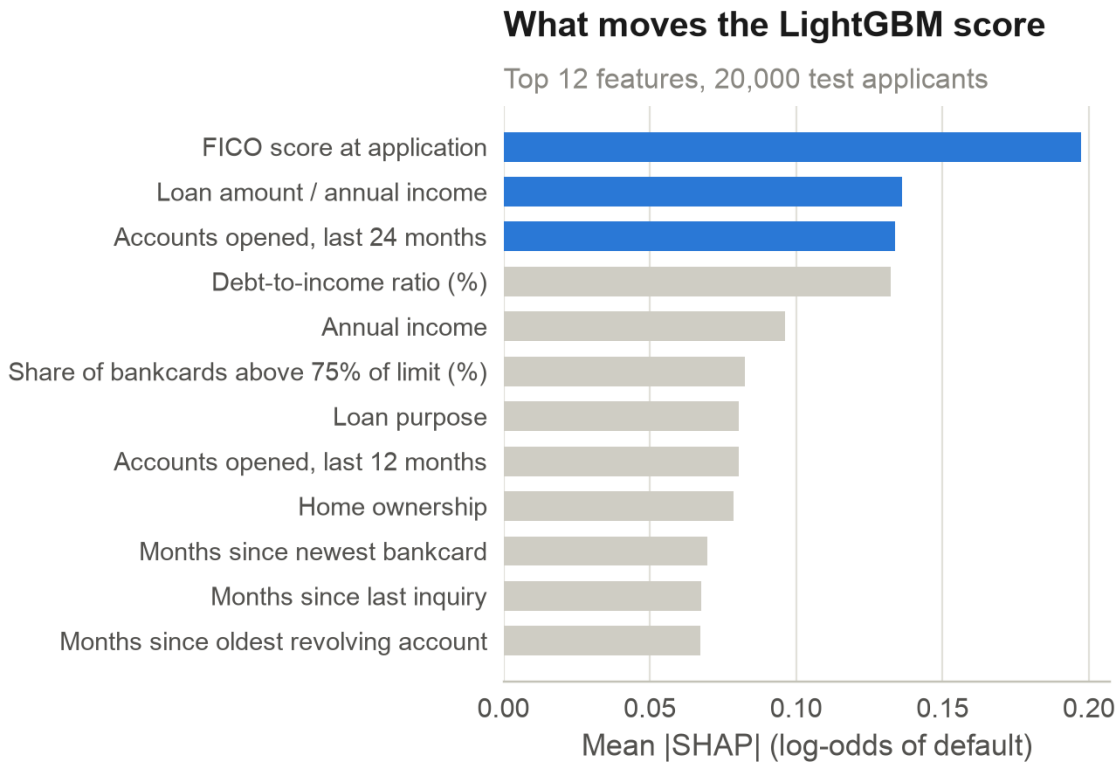
test AUC, vs 0.681 for LC sub-grade

- KS 0.29 vs 0.27 (grade), 0.25 (scorecard)
- Calibrated: mean PD 14.6%, observed 14.7%
- Stable: score PSI 0.005 train vs test

Test window: 161,997 loans issued Jul-Dec 2015

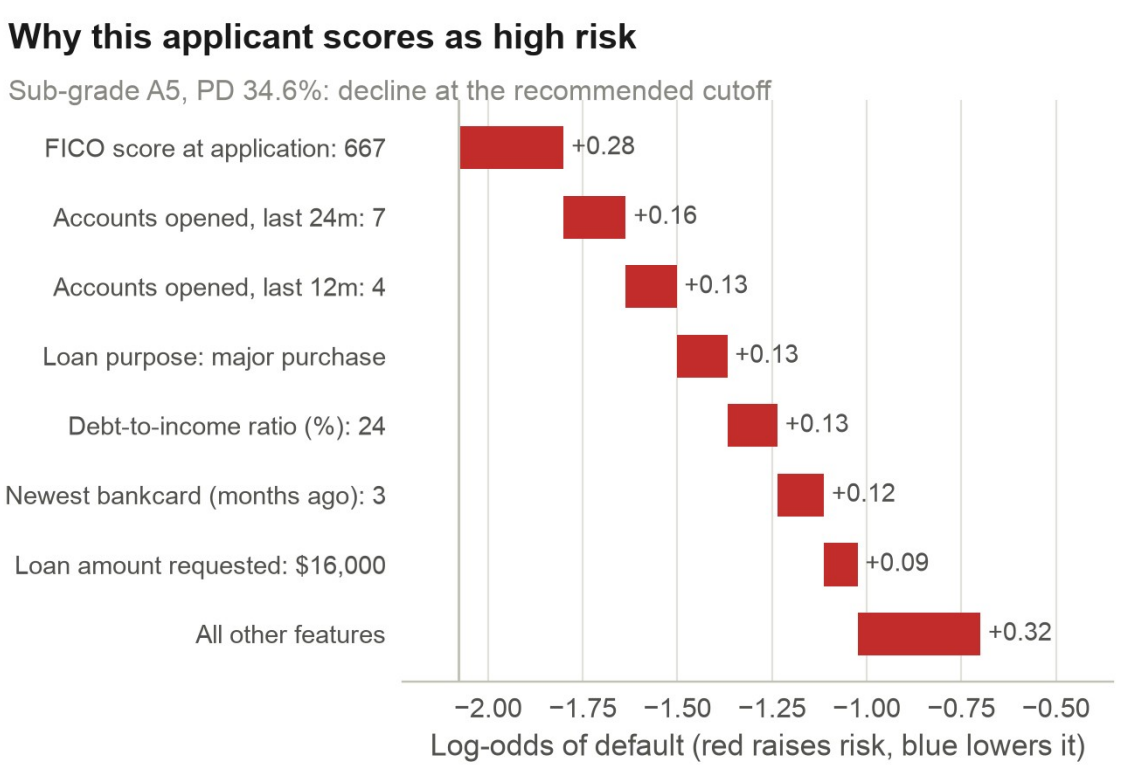
FINDING 4

FICO, loan size relative to income and recent credit-seeking drive the score



Test window sample, seed 42

Top 5 features carry 28% of the total SHAP effect.



Starting point = average applicant; test window, Jul-Dec 2015

A sub-grade A5 applicant declined: low FICO and recent new accounts.

RECOMMENDATION

Adopt the balanced cutoff: decline applicants with a LightGBM PD above 27.9%

CONSERVATIVE

Keep 82% of applicants

Decline PD above 22.7%. Default rate 11.5%. Per half-year vs today: profit +$18.8M, losses avoided $42.7M.

Dollar volume kept: 83.6%

RECOMMENDED

Balanced: keep 91% of applicants

Decline PD above 27.9%. Default rate 12.8%, from 14.7%. Per half-year: profit +$13.8M, losses avoided $26.0M.

Dollar volume kept: 91.4%

GROWTH

Keep 96% of applicants

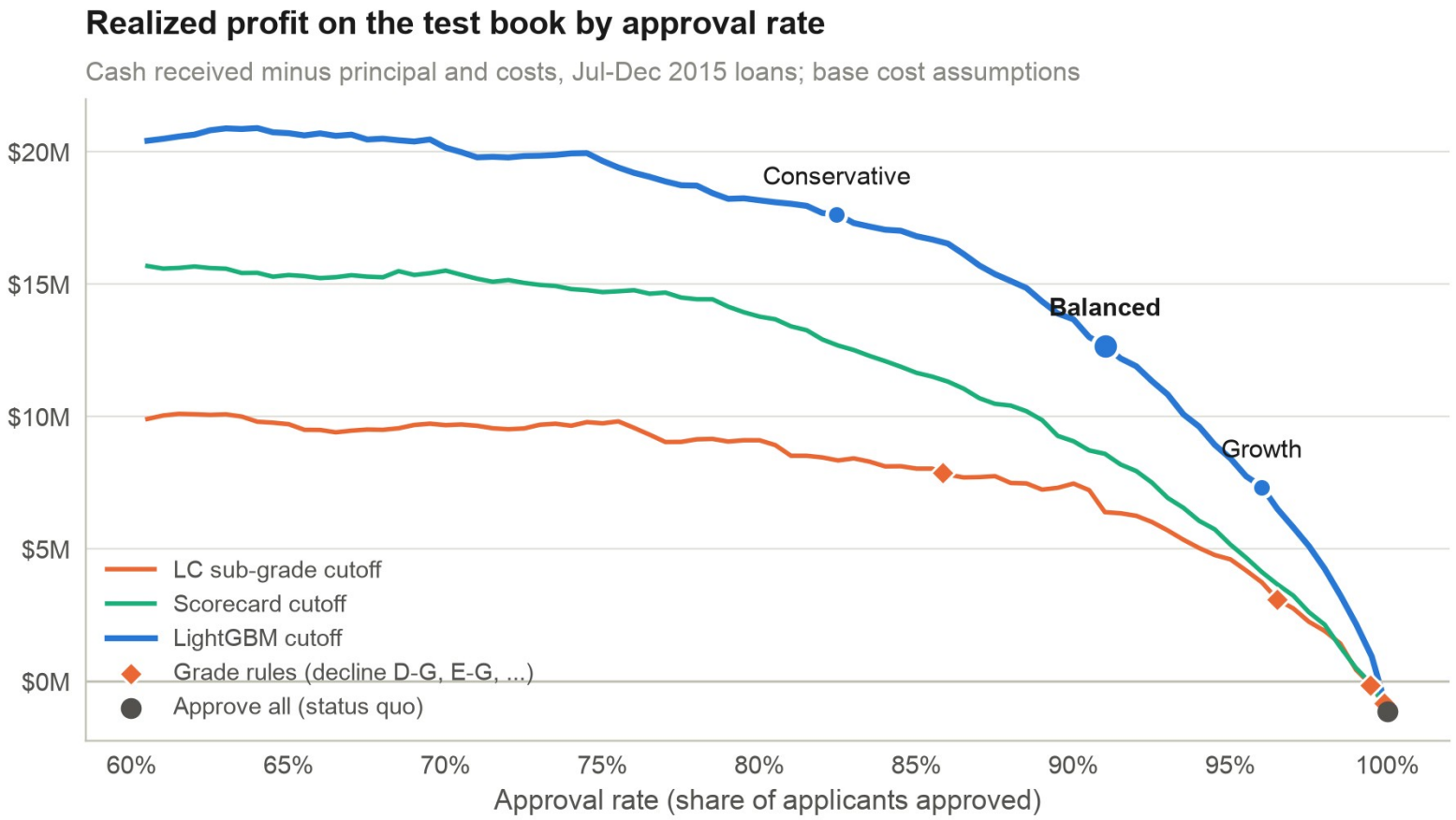
Decline PD above 33.3%. Default rate 13.7%. Per half-year vs today: profit +$8.5M, losses avoided $13.8M.

Dollar volume kept: 96.0%

Cutoffs chosen on Jan-Jun 2015 by expected profit; realized results on 161,997 Jul-Dec 2015 loans. Base cost assumptions.

EXPECTED IMPACT

The balanced cutoff turns a -$1.1M half-year into +$12.7M



Test window: 161,997 loans, $2.1B; cutoffs chosen on Jan-Jun 2015, never on test

+$27.6M

profit a year vs approving all, base assumptions

- $52.1M of credit losses avoided a year
- Same approval as declining D-G: $16.6M vs $7.9M
- Range +$21.2M to +$40.5M a year by cost scenario

Annual = 2 x Jul-Dec 2015 book ($4.2B a year)

LIMITATIONS AND NEXT STEPS

The policy can only tighten: Lending Club's rejected applicants are not in the data

Data limits

- Approved loans only: no reject inference
- FICO floor of 660: no subprime or thin-file view
- 2015 losses ran above training: 39.4% vs 35.9%
- Marketplace data used as a balance-sheet lender

Model limits

- AUC gain over the grade is modest: +0.019
- Pricing held fixed; no time value of money
- LightGBM needs SHAP; the scorecard is simpler to govern
- No fairness test on protected groups yet

Next steps

- Champion-challenger trial on live applications
- Re-price the 20%+ band: -$1.4M on test
- Monthly PSI, vintage curves and override rates
- Add 60-month loans once vintages resolve

APPENDIX

Methodology

Target and sample

Default = Charged Off or Default; good = Fully Paid. 589,249 individual 36-month loans issued 2012-2015; 147 unresolved and 239 joint applications excluded.

Split and leakage

Train 2012-2014 (306k), validation Jan-Jun 2015 (121k), test Jul-Dec 2015 (162k). 63 application features kept out of 151 columns.

Models

Scorecard: optbinning WoE, IV >= 0.02, |r| < 0.7, sign checks, 13 characteristics, 600 points at 50:1, PDO 20. LightGBM: 15 leaves, early stopping on validation.

Calibration and metrics

Platt recalibration on validation for every model. Test metrics: AUC, Gini, KS, Brier, decile calibration and PSI of the score between training and test.

P&L

Expected profit = principal x [(1 - PD) x G - PD x L - costs]. G by rate band and L = 35.9% from 2012-2014; costs = 4% funding x 1.30 years + 1% opex.

Policy choice

Cutoff maximises expected profit on Jan-Jun 2015 subject to approving at least 90% of applicants. Realized profit = cash received minus principal and costs on the test loans.